



İSTANBUL
GELİŞİM
ÜNİVERSİTESİ

Final Term Project Report

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Lesson Name: Internet and Web Programming

Lesson Number: SEN204

INTRODUCTION

Project Title:

BridgeBuddy — International Student Support and Survival Platform

Research Problem / Problem Statement:

International and Erasmus students moving to Türkiye often face major challenges related to language barriers, transportation systems, healthcare communication, accommodation, banking procedures, and adapting to a completely new cultural environment.

Most students depend on scattered information from social media groups, unofficial sources, or random online forums, which creates confusion and misinformation. Additionally, many students struggle to communicate in Turkish during emergencies or daily activities such as shopping, traveling, or visiting hospitals.

Traditional solutions do not provide a centralized and interactive platform specifically designed for international students. This project addresses the problem by developing a web-based support platform that provides survival guides, community discussions, and essential Turkish phrases to help students adapt more easily to life in Türkiye.

Objective of the Project:

The objective of this project is to develop a responsive and interactive web application using Django, HTML, CSS, Bootstrap, and SQLite database technologies.

The platform allows international students to:

- Access survival guides related to transportation, healthcare, banking, and residence permits
- Communicate and interact through a community discussion forum
- Learn essential Turkish phrases for daily communication and emergencies
- Search and filter useful information dynamically

- Share experiences and receive support from other students

The project also demonstrates backend development using Django framework, database management using models, user authentication systems, dynamic rendering, and responsive frontend design.

Expected Outcome:

The expected outcome of this project is a fully functional web platform that helps international students adapt more efficiently to life in Türkiye.

The system enables users to:

- Register and log into the platform
- Browse survival guides
- Search and filter guides dynamically
- Create community posts and comments
- Access Turkish language support phrases
- Interact with a responsive and user-friendly interface

Overall, the project demonstrates how modern web technologies can be used to solve real-world problems faced by international students.

METHODOLOGY

Approach to Problem Solving:

1. Control Flow of the System:

The system follows an event-driven and database-driven architecture where user interactions trigger backend processes dynamically.

When users interact with features such as login, searching guides, creating posts, or commenting, Django views process requests and update the database and user interface accordingly.

2. Input, Processing, Output:

Input:

- User registration and login information
- Community posts and comments
- Guide search keywords
- Category filter selections
- Turkish phrase management from admin panel

Processing:

- Django views handle HTTP requests
- Django models interact with SQLite database
- Authentication system validates users
- Search and filtering operations query database dynamically
- Templates render dynamic content using Django Template Language
- Bootstrap components improve responsiveness and UI presentation

Output:

- Display of guides and categories
- Dynamic community discussions
- User authentication and session handling
- Turkish phrase display
- Dashboard statistics
- Responsive user interface updates

3. System Control Flow Steps:

1. User opens the BridgeBuddy website
2. Homepage and navigation system are displayed
3. User explores guides, phrases, and community discussions
4. User registers or logs into the platform
5. User creates posts and comments
6. Django processes requests and stores data in database
7. Dynamic content updates appear on frontend
8. Users continue interacting with the system

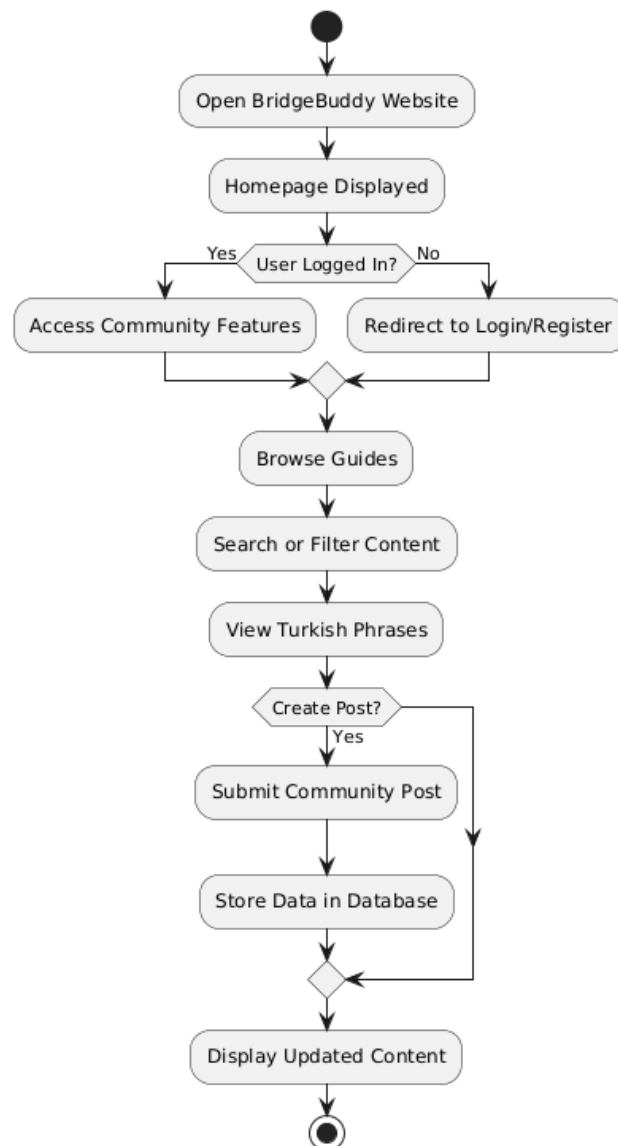


Figure 1: System Flowchart of BridgeBuddy

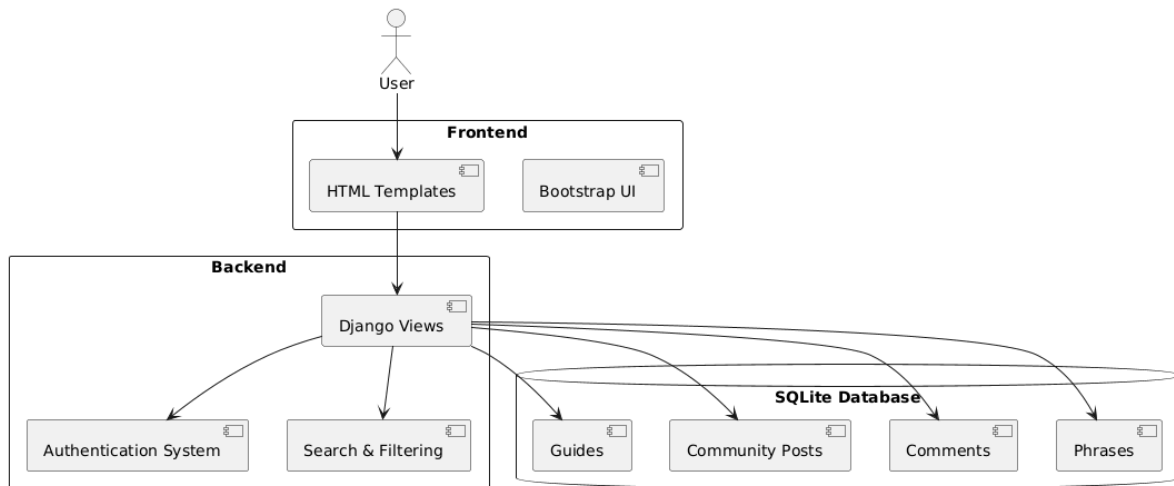


Figure 2: System Architecture and Component Interaction

TECHNICAL IMPLEMENTATION

Programming Languages:

- Python
- Django Framework
- HTML
- CSS
- Bootstrap
- SQLite Database
- JavaScript

Code Example:

```
@login_required
def create_post(request):

    if request.method == 'POST':

        title = request.POST.get('title')
        content = request.POST.get('content')

        CommunityPost.objects.create(
            user=request.user,
            title=title,
            content=content
        )

        return redirect('/community/')

    return render(request, 'main/create_post.html')
```

Explanation:

The project is implemented using Django framework for backend development and Bootstrap for frontend responsiveness and styling.

The system follows the Model-View-Template (MVT) architecture of Django:

- Models manage database structure and data handling
- Views process requests and business logic
- Templates render dynamic HTML pages

The authentication system allows users to register, log in, and interact securely with the platform.

Dynamic content such as guides, phrases, posts, and comments are fetched directly from the SQLite database using Django ORM queries.

Search and filtering functionality improve usability by allowing users to quickly locate relevant survival guides.

Additionally, responsive Bootstrap components ensure compatibility across different screen sizes and devices.

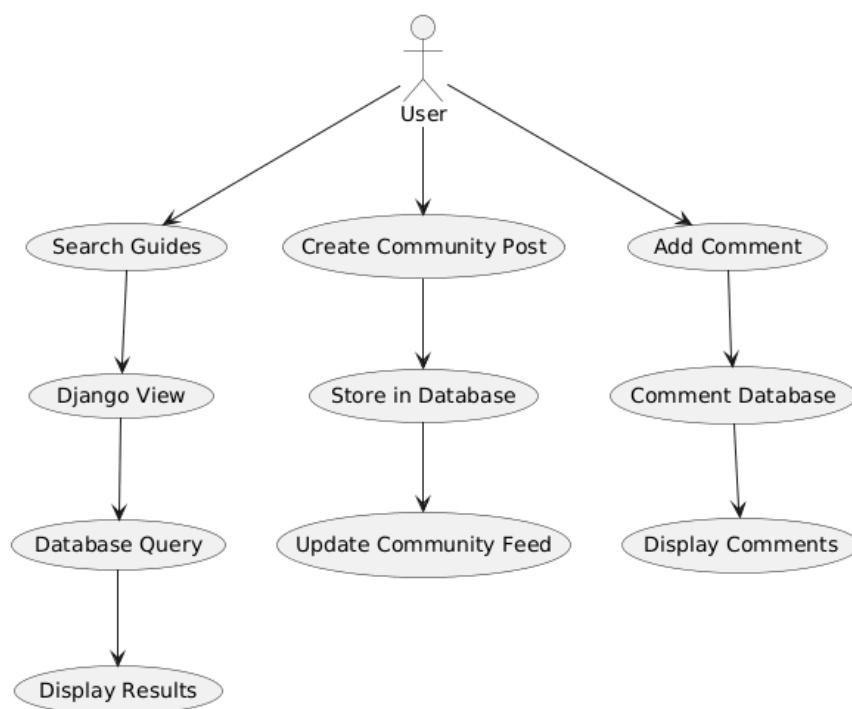


Figure 3: Data Flow in Community and Guide System

Conclusion:

BridgeBuddy successfully demonstrates how modern web technologies can be used to solve real-world challenges faced by international students.

The platform provides centralized access to survival guides, community support, and Turkish language assistance through an interactive and responsive web application.

The project also highlights important software engineering concepts such as authentication systems, database management, dynamic rendering, search functionality, and responsive UI design using Django framework.

Overall, the project provides both technical implementation experience and practical social impact by helping international students adapt more comfortably to life in Türkiye.